

# Nathan (Kaiyan) Zhang

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## Research Interests

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Causal inference and statistical learning for observational and high-dimensional data; Time series forecasting; Applied data science; Applications in economics, finance, labor, and text-rich data.

## Education

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### The University of British Columbia

Sep 2023 - May 2028

B.A. in Economics and Statistics; Minor in Data Science; Vancouver, BC

- Cumulative average: 93.5/100

## Research Experience

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### Research Assistant, UBC Sauder School of Business

Apr 2026 - Present

Labor, technology exposure, NLP, and empirical data construction

Supervisor: Dr. J. Frank Li and Dr. Cheng Chen

- Serve as the project's sole research assistant, owning the path from raw data cleaning to regression-ready samples; link CPS worker records, 60+ historical O\*NET task releases, occupation crosswalks, and USPTO/PATSTAT patent data with DuckDB and Polars.
- Run NER/RE on O\*NET task text to extract input-output task relations and summarize them into occupation-level task graphs; run transformer similarity jobs on UBC HPC covering about 56M PATSTAT patent records.
- Draft the manuscript data section and assemble an R replication package with GMM and fixed-effects models, summary tables, figures, and validation records.

### Research Assistant, UBC Sauder School of Business

May 2026 - Present

Financial text, source auditing, and investment-research data

Supervisor: Dr. Allen Hu

- Use yt-dlp to batch-download finance creator videos and audit metadata; organize creator snapshots, video tables, evidence joins, validation summaries, and runbook notes for reuse in finance-media research.
- Audit 10 TDM Studio financial-media metadata partitions for source coverage, missingness, and field reliability; export parquet datasets and run text diagnostics with word clouds, LDA topics, and Loughran-McDonald tone checks.

### Research Assistant, Vancouver School of Economics at UBC

Jan 2025 - May 2026

Econometrics, machine learning, and research-computing infrastructure

Supervisor: Dr. Jonathan Graves

- Developed reproducible Python and R workflows for panel econometrics, time-series forecasting, text embeddings, zero-shot classification, and computer-vision teaching modules.
- Maintained Quarto and GitHub Actions publishing pipelines for COMET and Praxis UBC materials, including online Jupyter image access for 1,000+ student users.
- Co-authored a Programming Historian tutorial under site-maintainer review on social-science NLI data preparation, including unstructured PDF parsing, citation removal, benchmark data construction, and validation examples.

## Selected Research Projects

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### Narrative Innovation Response Functions

In progress

Research project in progress with Jerrold Huang

Methods: financial text, event studies, semantic forecast errors, abnormal returns

- Co-designed the NIRF framework: predict expected disclosure from pre-event firm state, treat residual narrative content as narrative forecast error, and study how markets absorb unexpected text information. Built a WRDS S&P 500 2018-2025 event panel linking Capital IQ earnings-call transcripts, IBES EPS surprises, CRSP returns, and historical index constituents; matched 31k+ events and ran doubly orthogonal response curves, learner-invariance checks, placebo tests, and timing checks.

### Double Machine Learning for Payment-Friction Spending Effects

Nov 2025 - Dec 2025

Course research project with public report and code

Methods: Double ML, orthogonal scores, cross-fitting, CATE profiling

[GitHub](#) [Report](#)

- Used DoubleML on 5,583 respondents and 22,332 in-person person-days from the 2024 Atlanta Fed Diary of Consumer Payment Choice; after high-dimensional controls, estimated that lower-friction payment mixes were associated with about 5% higher daily spending. Built CATE profiles by ticket size, income, and education to locate settings where payment friction was more visible.

### Canadian Inflation Forecasting and Macro Signal Evaluation

Aug 2025

Course presentation and forecasting project

Methods: VAR, time series, rolling validation, forecast benchmarking

[GitHub](#)

- Built an R/Quarto VAR forecasting pipeline for Canadian CPI and macro-financial indicators; in rolling 12-month validation, reached 0.16% RMSE and used impulse responses to prepare a Bank of Canada-style policy presentation.

# Presentations and Workshops

Introduction to Computational Text Analysis for Research

Sep 10, 2025

APSA Migration and Citizenship Pre-Conference, UBC Centre for Migration Studies; Workshop materials contributor and technical facilitator

Workshop page

- Contributed demo, slides, and training materials for a hands-on computational text analysis workshop and assisted with in-session technical support.

# Teaching and Research Computing Materials

ECON 227 LLM Prediction Module

Co-author

Module

- Co-authored course-adopted materials connecting next-word prediction, token probabilities, and economics examples for students learning LLMs through applied research.

SOCI 280 Research Computing Materials

Research support, planning, editorial review, and maintenance

Module

- Supported material collection, planning, review, and maintenance for computational social science teaching materials.

Praxis UBC AI Literacy Modules

Contributor and maintainer

Praxis UBC

- Contributed public teaching modules on LLMs, embeddings, CNNs, convolutions, and research computing for non-CS student audiences.

# Teaching Experience

Teaching Assistant, UBC Chinese Language Program

2024 - 2025

Vancouver, BC

- Supported CHIN 131 and CHIN 133 through grading, classroom support, volunteer training, and feedback coordination.

# Awards and Scholarships

Work Learn International Undergraduate Research Awards (WLIURA) (UBC)

2026

Economics Undergraduate Scholarship (Vancouver School of Economics, UBC)

2025

Faculty of Arts International Student Scholarship (UBC Faculty of Arts)

2024, 2025

# Selected Coursework

**Statistical theory and methods:** STAT 302 - Introduction to Probability (91); STAT 305 - Introduction to Statistical Inference (100); STAT 406 - Methods for Statistical Learning (97); STAT 443 - Time Series and Forecasting (95)

**Mathematics and optimization:** MATH 220 - Mathematical Proof (100); MATH 307 - Applied Linear Algebra (100); CPSC 406 - Computational Optimization (92)

**Econometrics, causal inference, and machine learning:** ECON 398 - Introduction to Causal Methods (98); ECON 425 - Advanced Econometrics (98); CPSC 340 - Machine Learning and Data Mining (93)

**Economics theory:** ECON 304 - Honours Intermediate Microeconomics I (89); ECON 305 - Honours Intermediate Macroeconomics I (91); ECON 307 - Honours Intermediate Macroeconomics II (96)

**Registered coursework:** MATH 320 - Real Variables I; STAT 460 - Statistical Inference I; STAT 461 - Statistical Inference II; MATH 303 - Introduction to Stochastic Processes; CPSC 440 - Advanced Machine Learning; AI 360 - Deep Learning; CPSC 423 - Natural Language Processing; ECON 421 - Introduction to Game Theory and Applications

# Technical Skills

**Statistical computing:** Python, R, SQL, Julia, MATLAB, Jupyter, Quarto, Git/GitHub, HPC/Slurm

**Data and ML libraries:** pandas, NumPy, scikit-learn, PyTorch, Transformers, DuckDB, Polars, tidyverse, fixest, statsmodels

**Methods:** causal inference, statistical learning, time series forecasting, econometric modeling, text-as-data, reproducible workflows

# Languages

Mandarin Chinese: native; English: academic working proficiency; Japanese: JLPT N3 certified; currently around N2 level